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Environmental modulation of *Aspergillus*-induced post-harvest decay in *Musa paradisiaca*

Amrendra Kumar Singh*
University Department of Botany, Jai Prakash University, Chapra, Bihar, India

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Abstract- Post-harvest spoilage of banana (*Musa paradisiaca*) due to fungal pathogens significantly impacts fruit quality and marketability. The present study investigates the pathogenicity of selected *Aspergillus* species in relation to fruit maturity, temperature, and relative humidity. Three species, *Aspergillus niger, A. flavus,* and *A. fumigatus*, were isolated and identified through morphological and ITS-based molecular methods. Pathogenicity assays revealed that mature fruits exhibited rapid decay within 5 days, whereas pre-mature fruits showed delayed infection up to 8 days. Maximum pathogenicity was recorded at 25°C, particularly for *A. flavus*, while *A. fumigatus* demonstrated peak growth at 75% relative humidity. The findings indicate that both environmental factors and host maturity significantly influence fungal infection dynamics, emphasizing the importance of controlled storage conditions to reduce post-harvest losses.

Keywords: Post-harvest, *Musa paradisiaca, Aspergillus* species, pathogenicity, ITS-based molecular methods

INTRODUCTION

Banana (*Musa paradisiaca*) is one of the most important horticultural crops cultivated and consumed across tropical and subtropical regions, serving as a staple food and a major source of income for growers.[1] Its high nutritional value, including carbohydrates, vitamins, and essential minerals, makes it a key component of food security. The fruit exhibits a climacteric pattern of ripening marked by a rapid rise in respiration and ethylene production, which accelerates senescence after harvest.[2] High moisture content and soft tissue structure further increase its vulnerability to microbial invasion, leading to rapid deterioration during storage.

Fungal pathogens are a primary cause of post-harvest losses in banana, with species of *Aspergillus* frequently associated with fruit spoilage under storage conditions.[3] These fungi are widely distributed in the environment and possess strong adaptive capabilities that allow colonization of nutrient-rich substrates such as ripened fruits. Certain species are also known to produce mycotoxins, which reduce fruit quality and pose potential health concerns.[4]

Environmental conditions strongly influence fungal growth and infection dynamics. Temperature regulates metabolic activity, spore germination, and enzymatic processes, with most *Aspergillus* species showing optimal growth within the mesophilic range of 20–30°C.[5] Relative humidity determines water availability required for fungal metabolism and colonization. High humidity levels promote spore germination, surface colonization, and subsequent tissue degradation.[6] Host physiology is another critical factor affecting susceptibility to infection. Ripening in banana involves biochemical and structural modifications

*Corresponding author :
Phone : 9955006167
E-mail : amrkr007@gmail.com

such as increased sugar accumulation, degradation of cell wall components, and reduction in tissue firmness. These changes create favorable conditions for pathogen establishment and rapid disease development.[7] Mature fruits show greater susceptibility to fungal invasion compared to pre-mature fruits due to reduced structural resistance and increased nutrient availability.

A clear understanding of the interaction between host maturity, environmental conditions, and fungal pathogens is essential for improving post-harvest management practices. The present study evaluates the combined influence of fruit maturity, temperature, and relative humidity on the pathogenicity of selected *Aspergillus* species associated with banana fruits.

MATERIALS & METHODS

Sample Collection and Isolation

Pre-mature and mature banana (*Musa paradisiaca*) fruits were collected and used for the study. The fruits were first surface sterilized to remove any external contaminants. Small portions of infected tissue were then excised under aseptic conditions and placed on suitable culture media to promote fungal growth. The emerging fungal colonies were sub-cultured to obtain pure isolates, which were maintained for further analysis.

Identification of Fungal Species

The isolated fungi were identified through a combination of morphological, microscopic, and molecular approaches. Colony characteristics such as color, texture, and growth pattern were carefully examined on culture media. Microscopic observations were carried out using lactophenol cotton blue staining to study structural features like conidia and hyphae. Molecular identification was performed through amplification of the internal transcribed spacer (ITS) region using ITS1 and ITS4 primers. The amplified sequences were analyzed and compared with available database records to confirm the identity of the fungal species.

Pathogenicity Test

Healthy banana fruits were selected and used for pathogenicity assessment. Pre-mature fruits were inoculated with *Aspergillus niger*, while mature fruits were inoculated with *A. niger*, *A. flavus*, and *A. fumigatus*. The inoculated fruits were kept under controlled conditions for observation. Pre-mature fruits were monitored over a period of eight days, whereas mature fruits were observed for five days. Disease development was assessed based on visible symptoms such as discoloration, softening, and tissue breakdown.

Effect of Temperature

To evaluate the influence of temperature on fungal growth, inoculated fruits were incubated at two different temperature conditions, 25°C and 35°C. The progression of infection and the extent of tissue damage were recorded over time. Differences in fungal activity under these temperature conditions were compared to determine the optimal range for pathogenicity.

Effect of Relative Humidity

The effect of relative humidity on fungal growth was examined by maintaining inoculated fruits at different humidity levels, namely 30%, 50%, and 75%. Observations were made on the intensity of fungal growth and the degree of tissue degradation under each condition. Variations in infection severity across humidity levels were used to assess the role of moisture availability in disease development.

RESULTS

Identification of Pathogens

Three fungal species were successfully isolated from infected banana tissues and identified through morphological, microscopic, and molecular analyses. The isolates were confirmed as *Aspergillus niger*, *Aspergillus flavus*, and *Aspergillus fumigatus*. The identification was supported by colony characteristics, microscopic features, and ITS sequence comparison with reference database records.

Effect of Fruit Maturity

Clear differences in infection pattern were observed between pre-mature and mature banana fruits. Pre-mature fruits showed delayed symptom development, with visible signs of infection appearing after approximately eight days. The infection remained relatively restricted, mainly showing mild vascular discoloration.

Mature fruits showed rapid disease progression, with prominent symptoms developing within five days. The infected tissues exhibited extensive softening, discoloration, and breakdown, indicating a higher level of susceptibility compared to pre-mature fruits.(Table 1)

Effect of Temperature

Temperature had a noticeable influence on fungal pathogenicity. The highest level of infection was recorded at 25°C, where fungal growth was more active and disease symptoms developed more rapidly. Among the species

tested, *Aspergillus flavus* showed greater pathogenic activity under this condition.

At 35°C, fungal growth was still observed, though the rate of infection and severity of symptoms were comparatively lower. The results indicate that moderate temperature conditions favor fungal activity more effectively than higher temperature levels.(Table 2)

Effect of Relative Humidity

Relative humidity showed a strong influence on fungal growth and disease severity. At 30% humidity, fungal development was minimal, with very limited infection observed. At 50% humidity, moderate fungal growth occurred, accompanied by visible but less severe symptoms.

The highest level of pathogenicity was recorded at 75% relative humidity. Under this condition, rapid fungal colonization and extensive tissue degradation were observed. *Aspergillus fumigatus* showed particularly strong growth at higher humidity levels.(Table 3)

Table 1: Effect of Fruit Maturity on Disease Development in Banana

Fruit Type	Fungal Species Involved	Time for Symptom Appearance	Nature of Infection
Pre-mature	*Aspergillus niger*	8 days	Mild vascular discoloration, slow spread
Mature	*A. niger*, *A. flavus*, *A. fumigatus*	5 days	Rapid softening, tissue breakdown, severe decay

Table 2: Effect of Temperature on Fungal Pathogenicity

Temperature	Dominant Species	Infection Level	Observations
25°C	*Aspergillus flavus*	High	Rapid growth and pronounced tissue damage
35°C	Mixed species	Moderate	Slower infection, reduced severity

Table 3: Effect of Relative Humidity on Fungal Growth and Infection

Relative Humidity	Dominant Species	Growth Intensity	Observations
30%	None/ minimal	Low	Negligible infection
50%	Mixed species	Moderate	Noticeable but controlled decay
75%	*Aspergillus fumigatus*	High	Rapid colonization and severe tissue damage

DISCUSSION

The findings show a clear influence of fruit maturity on susceptibility to fungal infection. Mature banana fruits developed visible symptoms within five days, while pre-mature fruits required a longer period of about eight days. This variation can be explained by physiological changes during ripening. Accumulation of sugars, breakdown of cell wall components, and reduction in tissue firmness create conditions that favor rapid fungal colonization. Earlier studies have reported similar patterns where ripened fruits support faster pathogen growth due to reduced structural resistance and increased nutrient availability.[8,9]

Temperature also played a significant role in regulating fungal pathogenicity. The highest level of infection recorded at 25°C indicates that this temperature range supports optimal metabolic activity of *Aspergillus* species. Enzymatic processes involved in host tissue degradation are more active under such conditions, leading to faster disease development. A decline in infection intensity at 35°C suggests that higher temperature imposes stress on fungal growth, even though the organisms remain viable. Comparable findings have been reported where moderate temperature ranges enhanced fungal enzyme production and colonization efficiency.[10,11]

Relative humidity showed a strong effect on fungal growth and infection severity. Limited growth at 30% humidity and progressive increase at 50% and 75% highlight the importance of moisture availability. High humidity supports spore germination and facilitates penetration into host tissues, resulting in extensive decay. Studies on post-harvest pathogens have demonstrated that elevated humidity levels accelerate fungal proliferation and increase disease incidence in stored fruits.[12,13]

Differences among the fungal species were evident under varying environmental conditions. *Aspergillus flavus* showed higher activity at 25°C, indicating its adaptation to moderate temperature conditions. *Aspergillus fumigatus* exhibited enhanced growth at higher humidity levels, reflecting its preference for moisture-rich environments. *Aspergillus niger* was capable of infecting pre-mature fruits, suggesting a greater tolerance to less favorable host conditions. Similar species-specific responses have been described in studies focusing on fungal ecology and pathogenic behavior, where variations in enzymatic capacity and stress tolerance influence infection patterns.[13]

The interaction between host maturity and environmental factors determines the overall progression of disease. Faster decay in mature fruits under favorable temperature and humidity conditions indicates that these variables act together to influence infection dynamics.

These observations highlight the importance of maintaining appropriate environmental conditions during storage to minimize post-harvest losses.

CONCLUSION

The present study demonstrates that post-harvest decay in banana is strongly influenced by fruit maturity and environmental conditions. Mature fruits showed rapid infection within approximately five days, whereas pre-mature fruits exhibited delayed symptom development, extending up to eight days. This difference reflects the increased susceptibility of ripened tissues to fungal colonization. Temperature had a marked effect on pathogenicity, with maximum infection observed at 25°C. A reduction in infection intensity at 35°C indicates that moderate temperature conditions are more favorable for fungal growth and disease progression. Relative humidity also played a critical role, with minimal fungal activity recorded at 30% humidity and maximum growth observed at 75% humidity. The increase in infection severity with rising humidity highlights the importance of moisture availability in fungal development.

Species-specific variation was evident under different environmental conditions. *Aspergillus flavus* showed the highest pathogenicity at 25°C, while *Aspergillus fumigatus* exhibited maximum growth at higher humidity levels. *Aspergillus niger* was able to infect pre-mature fruits, indicating its ability to survive and colonize under relatively less favorable host conditions. These findings confirm that the interaction between host maturity, temperature, and relative humidity determines the extent of post-harvest decay in banana. Maintaining storage conditions below 25°C and relative humidity below 50% can significantly reduce fungal growth and delay spoilage. Careful handling of fruits at earlier maturity stages can further extend shelf life and minimize post-harvest losses.

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